



Complete Schedule of XXV ISSSD

September 22-26, 2025

Local Venue: CDTN - Belo Horizonte, Brazil

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XXV International Symposium on Solid State Dosimetry

22-26 SEPTEMBER 2025 - BELO HORIZONTE, BRAZIL



General Schedule

General Schedule					Hour		
Pre-Symposium - Monday, 22 Sep					8h		
Registration	Development of PET Radiopharmaceuticals - Dr. Marina Bicalho Silveira Dr. Isabel Dregely Dr. Emerson Bernardes Soares (Spectrum)		Introduction to Radiation Detectors - Dr. Marco A Polo Labarrios (Lattice)		9h		
					9h30		
					10h		
					10h30		
					11h		
COFFE BREAK					11h30		
LUNCH					12h		
					13h		
	Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico - Dr. Silvia S Hidalgo Tobón (Spectrum)		Solid State Dosimetry: Principles and Applications & OSL Advanced Materials - Dr. Ivonne Berenice Lozano Rojas Dr. Alida Tamar-Gabor (Lattice)		13h30		
					14h		
					14h30		
					15h		
					15h30		
COFFE BREAK					16h		
					16h30		
					17h		
Tuesday, 23 Sep		Wednesday, 24 Sep		Thursday, 25 Sep	Friday, 26 Sep	Hour	
Registration		Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	Invited Speaker: Luciana Carvalheira (Lumina)	Invited Speaker: Rolf Behrens (Lumina)	8h	
Opening Ceremony (Lumina)				Invited Speaker: Bernardo Maranhão Dantas Maria Jose Neves (Lumina)	Invited Speaker: Susana Lalic (Lumina)	8h40 9h	
COFFE BREAK				COFFE BREAK		Invited Speaker: Thiago Viana Miranda Lima (Lumina)	9h20
Invited Speaker: Wilson Calvo (Lumina)		Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	COFFE BREAK		10h	
Invited Speaker: Juan Azorin Nieto (Lumina)				Invited Speaker: Ademir Lugo (Lumina)	Oral Sessions (Lumina and Spectrum)	10h30	
				Invited Speaker: Richard Hugtenburg (Lumina)	Invited Speaker: Tarcisio Campos and Alberto Campos (Lumina)	11h 11h30	
LUNCH		LUNCH		LUNCH		12h30	
Oral Sessions (Lumina and Spectrum)	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Technical Visit	Oral Sessions (Virtual Room)	Invited Speaker: Helio Yoriyaz (Lumina)	Oral Sessions - (Spectrum and Lattice)	13h30	
	Roundtable: Radon in Environment: Tracer Applications and Public Health Risks (Lattice)			Invited Speaker: Georgia Joana (Lumina)		Oral Sessions (Lumina, Spectrum and Lattice)	14h
COFFE BREAK						Poster Session: 14h30 to 17h - Referees for evaluation - (Hall)	Invited Speaker: Yvone Maria Mascarenhas (Lumina)
	Roundtable: AI's challenges and Opportunities: Connecting research, industry and society (Lumina)	COFFE BREAK				COFFE BREAK	15h
	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Invited Speaker: Silvia Tobón (Lumina)	COFFE BREAK			15h30	
Welcome & Honors Night (Lumina)				Helen Khoury Mentorship WIN-Brasil (Spectrum)	Roundtable: Leadership in Scientific and Technological Research (Lumina)	16h	
					Invited Speaker: Carmen Bueno (Lumina)	16h30	
					Closing Session: Honor award & Helen Khoury award for best works (Lumina)	17h	
						17h30	
						18h00	

Monday, 09/22/2025

08:00 *Registration*

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Marina Bicalho	Chair: Álvaro M. L. Gómez
10:30 COFFE BREAK	08:00 Dr. Marina Bicalho, Dr. Isabel Dregely and Dr. Emerson Bernardes <i>Development of PET Radiopharmaceuticals</i>	08:00 Dr. Marco Antonio Polo La-barrios <i>Introdution to Radiation Detectors</i>

12:00 LUNCH

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Jhonny Benavente	Chair: Luiz Claudio Meira-Belo
15:00 COFFE BREAK	13:00 Dr. Silvia S Hidalgo Tobón <i>Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico</i>	13:00 Dr. Ivonne Berenice Lozano Rojas and Dr. Alida Tamar-Gabor <i>Solid State Dosimetry: Principles and Applications & OSL Advanced Materials</i>

Tuesday, 09/23/2025

08:00 *Registration*

Opening Cerimony - Lumina Auditorium

09:00 Opening Cerimony

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Silvia Hidalgo

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo,
Current Status of Strategic Projects at CNEN.

11:30 *Invited Speaker* – Juan Azorin Nieto,
Thermoluminescence and Its Application in Radiation Dosimetry.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Moderator: Talita Santos,
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology

Spectrum Auditorium
Chair: Tarcísio Campos

**Luminescent Materials/ Applications of
Thermoluminescence**
Lumina Auditorium
Chair: Álvaro M. L. Gómez

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Behavioral and Neuroinflammatory
Changes in the Hippocampus of Adult Mice Fol-
lowing Low-Dose Radiation Exposure During The
Neonatal Period*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spec-
trometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mol-
lusk Perna*

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Elec-
tronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerá-
micas de la Zona de Playa Vicente, Veracruz, México
Aplicando el Método de Datación Absoluta por Termo-
luminiscencia*

Roundtable Session (Lumina Auditorium)

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Moderator: Telma C. F. Fonseca,
Panelists: Rita Sebastião, Melina Castro and Ana Carolina

13:30 to 16:30 - Poster Session (Main Hall)

**Note: 14:30 to 16:30 - Referees for evaluation
In-Person**

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-001** - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLL-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
- G01-011** - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
- G01-026** - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
- G01-029** - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
- G01-032** - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
- G01-035** - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
- G01-036** - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
- G01-038** - Ketrin Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
- G01-040** - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
- G01-041** - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of CaSO₄-based radiation detectors*
- G01-044** - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha-beta – Pedra Branca State Park*
- G01-045** - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

- G05-002** - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
- G05-003** - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
- G05-006** - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
- G05-007** - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
- G05-015** - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ⁹⁰Y radioembolization: an intercomparison protocol for activity optimization*
- G05-017** - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
- G05-019** - Marcela Morais Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
- G05-020** - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*
- G05-022** - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

- G08-004** - Divanizia do Nascimento Souza, *Development of high-sensitivity CaSO₄:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
- G08-007** - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of CaMoO₄ powders*
- G08-008** - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*

G08-009 - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color types of quartz samples under 405 nm stimulation*

G08-010 - Rafael dos Santos Viana, *Promotion of the Compton effect of ^{137}Cs photons to enhance the efficiency of radioluminescent batteries*

G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*

G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free BaZrO_3 Perovskite Scintillators*

G08-014 - Claudete Roberta Evangelista, *Retrospective dosimetry implementation using automotive ceramic components*

G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $\text{CaSO}_4\text{:Yb,Mn}$ Phosphors*

G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*

G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Jhonny Benavente

16:00 *Invited Speaker* – Silvia S. Hidalgo Tobón,
State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes, Luíz Claudio Meira-Belo and Telma C. F. Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcisio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence
Spectrum Auditorium
Chair: Andrea Mantuano

08:00 Rafael Cogollo

G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α - Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya

G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes

G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos

G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho

G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

09:15 Paula Selvatice Pereira Teles

G01-031 - *Application of the MCNP code in dose verification for stereotactic radiosurgery (SRS)*

09:30 Alexander C. F. da Silva

G01-028 - *Absorbed Dose Evaluation in Breast Tissues Using Heterogeneous Models*

Dosimetry

Lattice Auditorium
Chair: Ana Carolina Costa

08:00 Berenice Guadalupe Olmos Grimaldo

G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Aline Fabiane Gonçalves de Oliveira

G02-012 - *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*

08:30 Renato Monaco Nunes

G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:45 Iasmin Vieira Nishibayaski

G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

09:00 Camilla da Silva Sampaio

G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:15 Laura Cardoso Takahashi

G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:30 Laisa de Almeida Neves

G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

10:00 COFFEE BREAK

Radiation Sources
Spectrum Auditorium
Chair: Teógenes da Silva

10:30 Roberto Méndez Villafañe

G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

10:45 Nayelly Nikole Díaz Gutiérrez

G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

Medical Physics
Lattice Auditorium
Chair: Lucas Paixão

10:30 Sara Suely Ventura da Silva Lima

G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla

G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior

G04-011 - *An AI revolution in radiotherapy in SUS:*

11:00 Roberto Méndez Villafañe
G06-003 - *Neutron measurements around the research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer*

11:15 Ismael Teixeira Vicente
G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:30 Estela Maria de Oliveira
G06-006 - *Comparative study of gamma efficiency curves in HPGe detector with different calibration standards*

11:45 Stanislav Svetachev
G06-007 - *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

12:00 Paulo Alberto Lima da Cruz
G06-010 - *Evaluation of thermal neutron-induced nuclear reactions in niobium foil by HPGe spectrometry and FLUKA/FLAIR Monte Carlo simulation*

The example of Mário Kroeff Hospital

11:30 Luis Alberto Perez-Martinez
G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

11:45 Teodoro Córdova Fraga
G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

12:00 Natasha Briggs e Silva
G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*

G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*

G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*

G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*

G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*

G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*

G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*

G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (HVL) in Computed Tomography Equipment*

G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*

G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*

G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*

G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main*

Parameters for Application in a Neutron Activation Analysis Laboratory

G02-035 - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*

G02-043 - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*

G02-044 - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*

G02-045 - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*

G02-046 - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*

G02-047 - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*

G02-048 - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*

G02-050 - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*

G02-051 - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*

G02-052 - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*

G02-061 - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*

G02-066 - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*

G02-070 - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*

G02-071 - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*

G02-072 - Mateus de Souza Resende, *NanoPapain with ^{99m}Tc : in Silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data of ^{99m}Tc -P-Nps*

G02-074 - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*

G02-077 - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*

G02-078 - Anderson Manoel Bezerra da Silva, *Optimization of Mn^{2+} and Tb^{3+} Concentrations in $\text{CaSO}_4:\text{Mn}, \text{Tb}$ and Advanced Investigation of its TL/OSL Dosimetric Properties*

G02-079 - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*

G02-080 - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation

On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

G01-007 - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ^{252}Cf neutron source on the ISO-phantom at LPN-CIEMAT*

G01-010 - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*

G01-013 - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*

G01-017 - Diego Eduardo Calvo-Medina, *Characterization of Beryllium Oxide as TLD of Clinical Use*

G01-019 - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*

G01-021 - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*

- G01-022** - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
- G01-033** - Edwar Alonzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataka II, Pernambuco, Brasil*
- G01-034** - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-001** - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
- G02-002** - Ernesto Saucedo-Hernandez, *Determination by GEANT 4 the biological damage caused by hadrons and carbon ions using a borated water phantom*
- G02-004** - José Edgardo Arellano Hernández, *Estudio de la respuesta termoluminiscente de $\text{Na}_2\text{O}-\text{Y}_2\text{O}_3\text{-P}_2\text{O}_5\text{-SiO}_2$ irradiados con Electrones*
- G02-008** - Woody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
- G02-009** - Woody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
- G02-013** - Tiago da Silva Jornada, *Longitudinal Monitoring of Occupational Radiation Exposure in Healthcare Professionals: Profile of Critical Hospital Sectors and Guidance for Radiological Protection Measures*
- G02-015** - Jardel Lemos Thalhofer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
- G02-016** - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of $\text{CaSO}_4\text{:Tm,Li}$ via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
- G02-022** - Akemi Yagui, *Analysis of Radiation Doses in Pediatric X-Ray Examinations*
- G02-024** - Caio César Santos Valeriano, *Application of Molecular Docking for Structural Modeling of Molecules in simulations with TOPAS-nBio: A Benchmark Against Conventional Material Definitions*
- G02-025** - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 detector for ground-level cosmic radiation monitoring*
- G02-029** - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*
- G02-037** - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in *Allium cepa* using micronucleus frequency: a comparative study between Antarctic and control conditions*
- G02-038** - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*
- G02-039** - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I*
- G02-054** - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*
- G02-059** - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*
- G02-060** - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*
- G02-067** - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*
- G02-073** - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*
- G02-086** - Marcelo Luís dos Santos Filho, *Study of $\text{CaSO}_4\text{:Tm,Li}$ Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*
- G02-087** - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom*
- G02-093** - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

- G03-005** - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer as-*

sociated with Gold Nanoparticles.

G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*

G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*

G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*

G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*

G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*

G04-015 - Anna Luiza Fraga da Silveira, *Characterization of the new FX-PEO dosimetric formulation in radiotherapy beams*

G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*

G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*

G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modeling with Monte Carlo methods in MR-Linac radiotherapy*

G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*

G5 - Radiological Protection

G05-001 - Natalia Matos Orozco, *Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos*

G05-008 - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*

G05-012 - Sitti Yani, *Monte Carlo simulation of flux and secondary particle analysis of gadolinium, cadmium, boron, and lithium-based neutron radiation shielding materials*

G05-013 - Tiago Amaral de Lima, *Physicochemical, Radiometric, and Elemental Characterization of Commercial Integral Grape Juices by Gamma Spectrometry and X-ray Fluorescence*

G05-018 - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^6LiI (Eu) Detector at the Neutron Metrology Laboratory*

G6 - Radiation Sources

G06-005 - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques*

G06-008 - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*

G06-011 - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ -ray spectrometry*

G7 - Radiobiology

G07-007 - Emilio Romero Muñoz, *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

G07-010 - Raúl Ken Delgado Velázquez, *Magnetic stimulation as a promoting factor for germination and development in *Guazuma ulmifolia* L. seeds*

G07-014 - Yasmym Sarmento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*

G07-016 - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

G08-003 - Marcelo Luis dos Santos Filho, *Development of $\text{CaSO}_4\text{:Li}$ Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*

G08-005 - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*

G08-017 - Caroline Pereira dos Santos, *Uncovering the intrinsic nature of the ternary metallic alloys Cu-NiPd and Cu_2PPd_2 : A DFT study*

G08-019 - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*

G08-021 - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce ($\text{SrAl}_2\text{Si}_2\text{O}_8\text{:Ce}$) sintetizado por el método de reacción en estado sólido*

G08-022 - Rodrigo Martínez Baltezar, *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Bruno Gallotti	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Priscila Santana/Rodrigo Gadelha
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13:30 Sandrith M. Pérez Ramos
G01-003 - *Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis*

13:45 Katherine G. Delgado
G01-005 - *Dosis de Radiación en Radiología Dental en Estudiantes de la Carrera de Estomatología*

14:00 Nayely D. M. Ramirez
G01-008 - *Synthesis of Y_2O_3 nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry*

14:15 Abel Fernando Martínez Cruz
G01-009 - *Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with Gd^{3+} and Li^+ , Obtained by a Glycine-based Combustion Process*

14:30 Michael A. M. Sánchez
G01-012 - *Analysis of TL half-life Using a better approach to the Exponential Integral*

13:30 Caroline Pereira dos Santos
G08-018 - *Unraveling the physical properties of pure and lithium thulium ion-doped calcium sulfate*

13:45 Carina Oliva Torres Cortés
G08-023 - *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5 M*

14:00 Pedro R. González
G08-024 - *Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters*

14:15 Jessica Paola C. Fraga
G02-005 - *La termoluminiscencia del Grafito para su posible aplicación dosimétrica*

14:30 Marcelo F. da Silva Santos
G02-014 - *New Approaches in the Synthesis of $\text{CaSO}_4\text{:Tm,Li}$ Phosphors with Sugars as Chelating Agents for Dosimetry Applications*

14:45 André Luiz Tavares e Silva

13:30 Erika Patricia Azorín Vega
G04-005 - *Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of Radiopharmaceutical Induced Bystander Effect*

13:45 Leticia González Zamora
G04-010 - *Study of Prefractal Antennas for Multiband Radiofrequency Applications*

14:00 Maria L. Miranda Maciel
G04-014 - *Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses*

14:15 Elsa Bifano Pimenta
G04-022 - *Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT*

14:30 Estefania Reyes Soto
G04-002 - *Automated*

- 14:45** Hanin Alali
G01-014 - *Assessment of Health Impacts among Medical Radiation Workers in a Tertiary Care Hospital*
- 15:00** Berenice Candia-Rodríguez
G01-015 - *Caracterización termoluminiscente del óxido de aluminio (Al_2O_3 : C) irradiado con diferentes fuentes de radiación ionizante*
- 15:15** Radmila Sabitova
G01-016 - *Characterization of additional irradiation positions in the IVG.1M research reactor*
- 15:30** Mohammed Alkhorayef
G01-018 - *Development of a Web-Based Tool for Single Time-Point Absorbed Dosimetry Calculation Using the H $\ddot{a}$ nscheid Approximation*
- 15:45** E. Cruz-Zaragoza
G01-020 - *Optically -and thermally stimulated luminescence detection of irradiated oregano (*Lippia graveolens*) polymineral*
- 16:00** Umme Muslima
G01-023 - *Thermoluminescence Characterization of Natural Flake Graphite Under 6 MeV Electron Beam Exposure from Medical LINAC for Dosimetric Applications*
- 16:15** Gabriela Pontes Cardoso
G08-001 - *Cathodoluminescence behavior of PVDF films doped with $CuSO_4$ under high-dose gamma irradiation*
- 16:30** Júlia Carina Orfão Costa
G08-002 - *Characterizations of $SrMoO_4:Dy^{3+}$ powders synthesized by hydrothermal method*
- 16:45** Daniele da Silva Machado
G08-006 - *Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis*
- 17:00** Daniel Santos Rodrigues
G08-011 - *Properties of $CaSO_4:Eu,Mn$ for Luminescent*
- G01-025 - A Data Science Approach on the Study of Al_2O_3 Thermoluminescent Glow Curves Under Varying Heating Rates*
- 15:00** Vinícius R. A. A. Martins
G02-055 - *Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging*
- 15:15** Rayline Leite da Silva
G02-084 - *Standardized Protocol for the Analysis of Cutting Samples and Lithological Materials by Gamma Spectrometry*
- 15:30** Nadia Rodrigues dos Santos
G02-049 - *Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park*
- 15:45** Carolina Salinas Domján
G02-090 - *Assessing Water-Equivalence in Gram-Negative Bacteria-Hydrogel Radiation Dosimeters: A Comparative Analysis with Gram-Positive Bacterial Systems*
- 16:00** Rayline Leite da Silva
G07-017 - *Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters*
- 16:15** Roos S. de Freitas Dam
G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*
- 16:30** Arthur Reis Martins
G06-015 - *Production of Radioisotopes in Nuclear Microreactors for Medical, Industrial, and Scientific Applications*
- 16:45** Jardel Lemos Thalhoffer
G02-010 - *Assessment of the level of natural radioactivity in the region of the future Municipal Environmental Protection Area (APA) Ilha Verde, Muriqui, Mangaratiba - RJ*
- Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data*
- 14:45** Giulia Rita de Souza Faés
G04-035 - *Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency*
- 15:00** Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xyleneol gel and Monte Carlo simulations*
- 15:15** Hericka O. Kenup-Hernandes
G05-004 - *Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot*
- 15:30** Bruna Gomes Araújo
G05-009 - *Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM- 41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste*
- 15:45** Kuanynsh Samarkhanov
G05-016 - *Radiological Characterization and Dosimetric Assessment of Liquid Radioactive Waste from BN-350 Reactor*
- 16:00** Jonathan O. dos Santos
G06-017 - *Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation*
- 16:15** Fatima Castañeda Ortiz
G07-001 - *Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line*
- 16:30** Daniele G. Mesquita
G07-006 - *Gamma Radiation Modulates Genomic Maintenance Pathways in T-Lymphocytes: An Analysis of Clinical Doses*
- 16:45** Walewska Gouveia Silva
G07-012 - *Proposal To Study the*

Dosimetry

17:00 Leonardo da Silva Boia
G02-036 - *Construction of the
Glioblastoma scenario in the right
optic nerve region using DIP
techniques for simulation in the
MCNP code with conversion to
lattice structure by SCAN2MCNP*

*Therapeutic Potential of the
Ionizing Effect of Low-Dose
Gamma Radiation in a Murine
Model of Alzheimer's Disease*

17:00 Francisco Torres Hoyos
G04-003 - *Breast carcinoma,
Fractals, Ballistic model,
Roughness, Scaling, Cluster*

13:30 to 16:00 TECHNICAL VISIT

Chair: Camila Engler

Places to visit at CDTN:

- Gamma Irradiation Laboratory - [More information, access here](#)
- Dosimeter Calibration Laboratory - [More information, access here](#)
- TRIGA IPR-R1 Reactor - [More information, access here](#)

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium

Chair: Bernardo Dantas

08:00 *Invited Speaker* – Luciana Carvalheira,

Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas,

Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves,

Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

14:30 - 17:00 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation

In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-081 - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*

G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*

G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*

G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*

G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

G03-001 - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*

G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*

G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*

G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*

G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*

G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*

G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

G04-006 - Otávio Gabriel Moreira Silva, *Estudo da Relação Sinal – Ruído em função do Índice de Ruído em exames de TC*

G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*

G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*

G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*

G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*

G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*

G04-033 - Roger Ferreira da Silva, *Mammography quality assurance based on phantom image with deep learning*

G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*

G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*

G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*

G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*

G07-009 - Olanía Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLL-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*

G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of Triple-Negative Breast Cancer*

G07-018 - Bruno Gallotti, *Toxicity of gold nanoparticles to yeast *Saccharomyces cerevisiae* BY4741 and UFMG 905 and radiosensitizing effect in low-energy X-ray beam*

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

10:30 Invited Speaker – Ademir Lugo, *Use of ionizing radiation for the production of nanoparticles and advanced curatives.*

11:30 Invited Speaker – Richard Hugtenburg, *New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.*

12:30 LUNCH

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

13:30 Invited Speaker – Hélio Yoriyaz, *Proton Beam Therapy: Main Characteristics and Challenges.*

13:30 Leonardo Santiago Melgaço Silva
G02-041 - *Development of an optical sensor prototype for dose readings in Fricke gel dosimeters*

13:45 Divanizia do Nascimento Souza
G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

14:00 Francisco Harley Dantas Hauradou Xavier
G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:15 Lucas Dias Rodrigues
G02-058 - *Impact of Granulometry on the Natural Radioactivity of Granite: Health Implications and Applications in Building Materials*

14:30 Francisco H. D. Hauradou Xavier
G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:45 José Elizeu de Souza Junior
G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

15:00 Raphael Francisco Gomes dos Santos
G02-063 - *Influence of Radiation Shielding on Effective Dose in Prostate Cancer Radiotherapy with a Varian Clinac 15 MV*

13:30 Ana Beatriz Caldeira de Andrade
G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using CaSO₄:Dy Dosimetry*

14:00 Nathalia Luzia Aidar Alves
G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

Invited Speaker - Lumina Auditorium

Chair: Andrea Mantuano

14:30 *Invited Speaker* – Georgia Joana,
Win Brasil - Gestão Diversidade & Inclusão.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WIN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Ketrim souza,
Panelists: Georgia Joana and Winners (Winbrasil)

16:00 Leadership in Scientific and Technological Research

Moderator: Luciana Carvalheira,

Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

**Radiation Protection
Lumina Auditorium
Chair: Rita de Cássia**

**Radiation Sources / Radiation Protection
Lattice Auditorium
Chair: Maria da Penha**

17:00 Raphael Francisco Gomes dos Santos

G05-005 - *Determination of Radiation Protection*

Procedures for an Offshore Plant Inspection in Brazil

Using Radiotracer Techniques

17:00 Gabriel do Nascimento Freitas

G06-014 - *Nuclear simulation workflow to correct*

gamma ray well logs for radioactive contaminated

drilling fluids

17:15 Mirella Maturo da Cruz

G05-021 - *Transfer Factors of Natural Radionuclides*

in Organic Foods from Rio de Janeiro, Brazil

17:15 José Alípio dos Santos Filho

G06-016 - *Proposal for the construction of a neutron*

generator for application in research and industry

17:30 Elydio J. D. Soares

G05-010 - *Investigation of radon exhalation from*

mortars made from niobium tailings and the

mitigating role of coating materials

17:30 Welen Nunes de Lima

G05-014 - *Proposal for the use of thorium as fuel in*

TRIGA[®] research reactors

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

08:00 *Invited Speaker* – Rolf Behrens,
From dosimeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic,
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima,
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

**Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno**

**Dosimetry
Lumina Auditorium
Chair: Peterson Squair**

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ -ray Spectrometry and Algorithmic Data Processing*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

Invited Speaker - Lumina Auditorium

Chair: Bruno M. Mendes

11:30 *Invited Speaker* – Tarcisio P. R. de Campos and Alberto Mizrahy Campos,
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

**Dosimetry
Spectrum Auditorium
Chair: Priscila Santana**

**Dosimetry
Lattice Auditorium
Chair: Bruno M. Mendes**

**Applications of Thermoluminescence
Lumina Auditorium
Chair: Teogenes da Silva**

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard an Aircraft from Southeastern Brazil to the Antarctic Continent*

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

14:00 Rafael S. Oliveira

13:30 Tarcisio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban Quarries in Belo Horizonte, Brazil*

14:00 Paulo J. Iack Silva
G02-068 - Monte Carlo N-Particle
(MCNP) Simulations for
Scintillation Emission in
Quantum Dot-Doped Systems

G02-076 - On the adaptation of a
commercial Bonner Sphere
Spectrometer to be used with
thermoluminescent detectors

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

14:30 *Invited Speaker* – Yvone Maria Mascarenhas,
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean.

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

16:00 *Invited Speaker* – Carmen Bueno,
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry.

17:00 CLOSING SESSION

- Final Session: Summary and Perspectives: Telma C. F. Fonseca & Bruno M. Mendes
- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade